

Amit Te'eni

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Legal name: Amit Teeni

Education

- 2021 – 2026  **Ph.D. in Engineering, Bar-Ilan University** Supervisor: Prof. Eliahu Cohen
Thesis title: *Entropic and geometric methods in quantum information theory*
Average grade: 100
- 2020 – 2021  **M.Sc. in Mathematical and Theoretical Physics, University of Oxford**
Graduated with distinction
- 2017 – 2019  **M.Sc. in Mechanical Engineering, Ben-Gurion University of the Negev**
Supervisor: Prof. Avishy Carmi. Average grade: 96.89 (summa cum laude)
Thesis title: *Novel Bell-type inequalities and their implication for quantum nonlocality*
- 2012 – 2017  **B.Sc. in Physics and Computer Science (double-degree track), Technion – Israel Institute of Technology.** Average grade: 90.9 (cum laude)
Final project title: *Pilot contamination mitigation using graph theory*

Fellowships and awards

- 2024  **Prof. Rahamimoff Travel Grants for Young Scientists**, US-Israel Binational Science Foundation (BSF)
- 2022 – 2025  **Scholarship for Outstanding PhD Students in Quantum Science and Technology**, Israeli Planning and Budgeting Committee
- 2021  **INL and BINA Joint Call for Collaborative Research**
- 2021 – 2025  **Bar-Ilan University Presidential Doctoral Fellowship of Excellence**
- 2021 – 2023  **Nano Scholarship**, Bar-Ilan Institute of Nanotechnology & Advanced Materials (BINA)
- 2020  **Oxford-Reuben Foundation Graduate Scholarship**

Conferences and seminars

Talks

- 2/2026  **One-query quantum algorithms for the index- q hidden subgroup problem**, QERNEL General Meeting 2026, Tel Aviv University
- 11/2025  **Equivalence of mutually unbiased bases via orbits: general theory and a $d = 4$ case study**, Atomic and Laser Physics seminar, University of Oxford
- 7/2025  **Oracle Problems as Communication Tasks and Optimization of Quantum Algorithms**, The 70th Annual Meeting of the Israel Physical Society, Technion – Israel Institute of Technology
- 4/2025  **Oracle Problems as Communication Tasks and Optimization of Quantum Algorithms**, QERNEL Launch Event, Bar-Ilan University
- 2/2025  **Mathematical Origins of Quantum Computational Speedup**, International Centre for Theory of Quantum Technologies (ICTQT), University of Gdańsk
- 3/2024  **Mathematical Origins of Quantum Computational Speedup**, Quantum and Linear-Optical Computation seminar, Iberian International Nanotechnology Laboratory (INL)
- 2/2020  **Correlation Minor Norm as a Detector and Quantifier of Entanglement**, The 65th Annual Meeting of the Israel Physical Society, Weizmann Institute of Science
- 9/2019  **Multiplicative Bell Inequalities**, Quantum Foundations seminar, Perimeter Institute for Theoretical Physics

Conferences and seminars (continued)

- 7/2018  **Multiplicative Bell Inequalities**, International Conference for New Foundations in Physics (ICNFP), Orthodox Academy of Crete

Posters

- 1/2025  **Oracle Problems as Communication Tasks and Optimization of Quantum Algorithms**, *Quantum SpeedUp* workshop, International Centre for Theory of Quantum Technologies (ICTQT), University of Gdańsk
- 2/2022  **Correlation Minor Norm as a Detector and Quantifier of Entanglement**, The Israel Physical Society Conference, Ben-Gurion University of the Negev
- 9/2018  **Multiplicative Bell Inequalities**, Conference for Young Quantum Information Scientists (YQIS), University of Vienna

Academic visits

- 2/2025  **International Centre for Theory of Quantum Technologies (ICTQT), University of Gdańsk, Poland**
A week-long visit to the New Quantum Resources group led by Prof. Pawel Horodecki
- 8/2024-10/2024  **University of California, Davis, United States of America**
A two-month-long visit to work with Prof. Greg Kuperberg, funded by a US-Israel Binational Science Foundation (BSF) Prof. Rahamimoff Travel Grant
- 3/2024  **International Iberian Nanotechnology Laboratory (INL), Braga, Portugal**
A short visit to the Quantum and Linear-Optical Computation group led by Prof. Ernesto Galvão
- 8/2019  **University of Ottawa, Ontario, Canada.** A month-long visit to the Structural Quantum Optics (SQO) group led by Prof. Ebrahim Karimi

Work experience

Undergraduate Final Project Supervisor, Bar-Ilan University, Ramat-Gan, Israel

-  **Characterization and Optimization of Quantum Algorithms**, 2024/25
-  **Multipartite Entanglement Detection via Correlation Minor Norms**, 2021/22

Teaching Assistant, Bar-Ilan University, Ramat-Gan, Israel

-  **Linear Algebra**, 2021/22 Semester A, 2022/23 Semester A
-  **Applied Thermodynamics**, 2021/22 Semester B, 2022/23 Semester B, 2023/24 Semester B
-  **Functional Analysis**, 2023/24 Semester B, 2024/25 Semester B

Part-time Lecturer, Department of Computer Science, Holon Institute of Technology, Israel

-  **Data Structures**, 2019/20 Semester B
-  **Operating Systems**, 2019/20 Semester B

Teaching Assistant, Department of Mechanical Engineering, Ben-Gurion University of the Negev

-  **Introduction to Computer Science**, 2017/18 Semester B, 2018/19 Semester B
-  **Undergraduate Laboratory in Mechanical Engineering (lab instructor)**, 2017/18 Semester A, 2018/19 Semester A